Antonio **Acuaviva Huertos**

Doctoral Researcher in Pure Mathematics · Lancaster University

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Education

Lancaster University

Lancaster, UK

PhD in Pure Mathematics

October 2023 – Present

- Expected PhD award: late April 2027.
- Supervisor: Professor Niels J. Laustsen.
- Research in functional analysis, with emphasis on Banach space theory and operator theory: complemented subspaces and primariness, $C(K)$-spaces, operator ideals and Calkin algebras, and nonlinear Banach space theory.
- Since summer 2026, systematic use of large language models in mathematical research; coauthor of *Mathematical discovery in the wild: AI-guided proofs in Banach space theory* ([17]).
- Awarded the **Pure Mathematics Prize** for exceptional performance during the first and third years of the PhD programme.
- EPSRC Doctoral Training Partnership Studentship (2023–present).

University of Cambridge

Cambridge, UK

Master of Advanced Study in Mathematics (Part III)

October 2022 – June 2023

- Result: **Distinction**.
- Received the **1912 Senior Scholarship** and **Fitzwilliam College Prize** for academic performance.
- Supported by the ***la Caixa* Foundation Postgraduate Fellowship Abroad**.

Universidad Complutense de Madrid (UCM)

Madrid, Spain

Double Degree in Mathematics and Physics

September 2017 – June 2022

- Ranked **1st of 171 graduates in Mathematics** and **1st of 247 graduates in Physics**.
- Recipient of the **Bachelor's Degree Extraordinary Award** in both Mathematics and Physics.
- Awarded the **Arturo Duperier Award** for the best academic record in Physics at UCM.
- Received **Community of Madrid Excellence Grants** in four consecutive years.
- Physics thesis: *Resilient Source Seeking with Robot Swarms*. Published summary: arXiv:2408.10152.

University of California, Berkeley

Berkeley, USA

Non-degree exchange programme (UCEAP)

August 2019 – June 2020

- Selected for a year-long study program; recipient of the **UCM–University of California Scholarship**, awarded to the **top seven students** at Universidad Complutense de Madrid.

Publications

Published or Accepted Journal Articles in Analysis

[1] **A. Acuaviva**. *The class of Banach lattices is not primary*. *Forum of Mathematics, Sigma*, 14:e41, 1–18 (2026).

[2] **A. Acuaviva** and A. B. Nasseri. *Quantifying (non-)weak compactness of operators on AL- and $C(K)$-spaces*. *Proceedings of the Edinburgh Mathematical Society*, 69(3):1088–1143 (2026).

[3] **A. Acuaviva**. *Complemented subspaces of a $C(K)$-space constructed by Candido*. *Journal of Mathematical Analysis and Applications*, 556(1):130115 (2026) (11 pages).

[4] **A. Acuaviva**. *Operators on injective tensor products of separable Banach spaces and spaces with few operators*. *Journal of Functional Analysis*, 290(1):111203 (2026) (17 pages).

[5] **A. Acuaviva**. *Factorizations and minimality of the Calkin algebra norm for $C(K)$-spaces*. *Journal of the London Mathematical Society*, 112(4):e70321 (2025) (20 pages).

[6] **A. Acuaviva** and D. Seco. *Equidistribution of zeros of some polynomials related to cyclic functions*. *Analysis and Mathematical Physics*, 12(2):53 (2022) (24 pages).

[7] **A. Acuaviva**. *$C(K)$-spaces with few operators relative to posets*. *Topology and its Applications*, article 109987 (2026). Available online 2 October 2026 (journal pre-proof).

Publications (continued)

Other Published or Accepted Journal Articles

[8] **A. Acuaviva**, J. Esson, E. Kastis, and B. Schulze. *Curved RUM spectra of periodic frameworks*. *Bulletin of the London Mathematical Society*, 58(10):e70513 (2026).

[9] J. Bautista[*], **A. Acuaviva**[*], J. Hinojosa, W. Yao, J. Jimenez, and H. Garcia de Marina. *Fully distributed and resilient source seeking for robot swarms*. *IEEE Transactions on Automatic Control*, Early Access, pp. 1–16 (2026). [*]Equal contribution.

[10] **A. Acuaviva**, A. Chan, S. Eldridge, C. Godsil, M. How-Chun-Lun, C. Tamon, E. Wright, and X. Zhang. *State transfer in chiral quantum walks*. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 383(2306):20240420 (2025) (19 pages).

Conference Proceedings

[11] **A. Acuaviva**, J. Bautista, W. Yao, J. Jimenez, and H. Garcia de Marina. *Resilient source seeking with robot swarms*. In *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pp. 57–63 (2024).

Preprints in Functional Analysis

[12] **A. Acuaviva**. *A solution to Michael's problem for Fréchet algebras*. arXiv:2609.28584 (2026) (8 pages).

[13] **A. Acuaviva**. *On complemented subspaces of* $L_1[0,1]$. arXiv:2609.17283 (2026) (12 pages).

[14] **A. Acuaviva**. *A negative solution to the complemented subspace problem for Banach spaces with unconditional bases*. arXiv:2609.08921 (2026) (26 pages).

[15] **A. Acuaviva** and T. Kania. *Banach's isometric conjecture over the complex field*. arXiv:2608.18257 (2026) (17 pages).

[16] **A. Acuaviva** and P. Acuaviva. *The uniform primary factorisation property for* $C(K,E)$. arXiv:2609.18993 (2026) (16 pages).

[17] **A. Acuaviva** and P. Acuaviva. *Mathematical discovery in the wild: AI-guided proofs in Banach space theory*. arXiv:2607.17388 (2026) (141 pages).

[18] **A. Acuaviva** and P. Acuaviva. *A unital Banach algebra which is not a Calkin algebra*. arXiv:2607.15075 (2026) (21 pages).

[19] **A. Acuaviva**. *A separable Banach space with a Schauder basis which is not a Lipschitz retract of its bidual*. arXiv:2607.12935 (2026) (21 pages).

[20] **A. Acuaviva**, B. Horváth, and T. Kania. *Pure infiniteness and primary factorisation*. arXiv:2607.01467 (2026) (19 pages).

[21] **A. Acuaviva**. *Preservation of primariness under* ℓ_1*-,* c_0*-, and* ℓ_∞*-sums of Banach spaces*. arXiv:2606.26417 (2026) (39 pages).

[22] **A. Acuaviva**. *Primariness of the spaces* $\ell_p(C(K))$ *for* $1 \leq p \leq \infty$. arXiv:2605.29854 (2026) (35 pages).

[23] **A. Acuaviva** and T. Kania. *Primariness and the primary factorisation property*. arXiv:2605.21711 (2026) (74 pages).

Other Preprints and Manuscripts

[24] **A. Acuaviva**, A. Acuaviva, and P. Acuaviva.
CNOT-distance is NP-complete under all-to-all connectivity. arXiv:2608.03825 (2026) (14 pages).

Selected Research Experience

Iliad

Iliad Fellowship (forthcoming) *2 November 2026 – 5 February 2027*

Tokyo University of Science *Tokyo, Japan*

JSPS Short-Term Pre/Postdoctoral Fellowship for Research in Japan (forthcoming)

Fields Institute for Research in Mathematical Sciences *Toronto, Canada*

Undergraduate Researcher, Fields Undergraduate Summer Research Program *June 2021 – August 2021*

- Investigated state transfer in quantum walks with Ada Chan, Christino Tamon, and Xiaohong Zhang; the project contributed to the later journal article *State Transfer in Chiral Quantum Walks*.
- Presented *Quantum Walks on Graphs: Perfect State Transfer and Periodicity* at the Canadian Undergraduate Mathematics Conference (2021).

Institute of Mathematical Sciences (ICMAT) *Madrid, Spain*

Introduction to Research Grant Recipient *June – September 2020 & 2021*

- Received the competitive Severo Ochoa Introduction to Research Grant in two consecutive years and undertook projects in mathematical analysis.

Employment

Lancaster University *Lancaster, UK*

Graduate Teaching Assistant (GTA) *October 2023 – December 2025*

- Taught on Numbers and Relations (MATH111), Discrete Mathematics (MATH112), Real Analysis (MATH210), and Complex Analysis (MATH215), including marking, tutoring, and problem classes.

Professional Service

Referee for *Studia Mathematica* and the *Banach Journal of Mathematical Analysis*.

Honours & Awards

2024, 2026	**Pure Mathematics Prize**, Lancaster University
2025	**Spring Into Quant Finance Programme**, G-Research
2023	**EPSRC Doctoral Training Partnership Studentship**, Lancaster University
2023	**1912 Senior Scholarship**, Fitzwilliam College, University of Cambridge
2023	**Fitzwilliam College Prize**, Fitzwilliam College, University of Cambridge
2022	**Postgraduate Fellowship Abroad**, *la Caixa* Foundation
2022	**Bachelor's Degree Extraordinary Award**, Universidad Complutense de Madrid (Ranked 1st in class)
2022	**Department Research Grant**, Dept. of Mathematical Analysis, UCM
2022	**Blas Cabrera Summer Scholarship**, Universidad Internacional Menéndez Pelayo and Spanish Ministry of Science and Innovation
2022	**Arturo Duperier Award**, Pedro Bernardo City Council (best academic record in Physics at UCM)
2018–2021	**Excellence Grants**, Community of Madrid (awarded in four consecutive years)
2020	**Becas Atrévete 2020**, McKinsey & Company
2019	**UCM–University of California Scholarship**, Universidad Complutense de Madrid (top seven students selected)
2019	**Santander Progress Scholarship**, Santander Bank
2015–2017	**Goñi and Rey Foundation Fellowship**, Full academic funding

Languages

Languages English (Full Professional Proficiency), Spanish (Native), German (basic)

Talks

Invited Seminar Talks

19 November 2026 **Invited seminar talk (forthcoming)**
Séminaire d'Analyse Fonctionnelle, Paris, France.

28 October 2026 **Invited seminar talk (forthcoming)**
Banach and Metric Space Geometry Seminar, Texas A&M University. Online.

20 May 2026 **The class of Banach lattices is not primary**
Seminar on Geometry of Banach Spaces, Jagiellonian University, Kraków, Poland.

14 May 2026 **Banach space constructions from almost disjoint families**
Working Group in Applications of Set Theory, Institute of Mathematics, Polish Academy of Sciences (IMPAN), Warsaw, Poland.

13 May 2026 **$C(K)$-spaces with few operators relative to posets**
Topology and Set Theory Seminar, University of Warsaw, Warsaw, Poland.

12 May 2026 **The class of Banach lattices is not primary**
Functional Analysis Seminar, IMPAN, Warsaw, Poland.

Conference & Workshop Talks

26 August 2026 **Quantitative (Non-)Weak Compactness of Operators on L_1- and $C(K)$-Spaces**
Brazilian Workshop on Banach Spaces (BWB 2026), Ubatuba, Brazil.

23–25 March 2026 **The class of Banach lattices is not primary**
Young Functional Analysts' Workshop, Cardiff University, Cardiff, UK.

2–4 April 2025 **Factorizations and norms on Calkin algebras of $C(K)$-spaces**
Young Functional Analysts' Workshop, University of Glasgow, Glasgow, UK.

7–12 July 2024 **Factorizations and norms on Calkin algebras of $C(K)$-spaces**
Banach Algebras and Operator Algebras, University of Waterloo, Waterloo, Canada.

3–6 July 2024 **Factorizations and norms on Calkin algebras of $C(K)$-spaces**
BAFALASA, Lancaster University, Lancaster, UK.

2021 **Quantum Walks on Graphs: Perfect State Transfer and Periodicity**
Canadian Undergraduate Mathematics Conference (CUMC). Remote.